# Putting the Circle Theorems Together
## Answer Key

High School Geometry

### Quick Answers

1. 43
2. 74
3. 6.28 cm
4. 15
5. (a) radius = 7 m, (b) y-coordinate of the tangent point = 5
6. (a) x = 25, (b) measure of angle A = 65
7. 150.80 $\text{cm}^2$
8. 18
9. (a) arc length = 12.57 cm, (b) sector area = 62.83 $\text{cm}^2$
10. (a) centre-radius form = $(x-5)^2 + (y+2)^2 - 100$, (a) radius = 10, (b) distance from centre to the chord = 6, —

### What is verified

15 of 16 answers machine-checked against SymPy, across 10 problems.
— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

### Curriculum

**Level:** High School Geometry
HSG-C.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
*Difficulty 1–5 of 5 across 16 tagged checks.*

### Common wrong answers

7. *If they got 25.13: used the arc-length formula instead of the sector-area formula*
8. *If they got 1.5: used the tangent length itself instead of its square*

**1.** An inscribed angle intercepts an arc of 86°. Find the inscribed angle.

**Solution:** The inscribed angle theorem: an inscribed angle is half the arc it intercepts. So the angle is 86°/2. $\boxed{43°}$

**2.** Inscribed angle $\angle ABC = 37°$ intercepts arc $AC$. Find arc $AC$.

**Solution:** Same theorem, read backwards: the arc is twice the inscribed angle, $2 \times 37°$. $\boxed{74°}$

**3.** Radius 9 cm, central angle 40°. Find the arc length.
**Solution:** An arc is the fraction $\frac{40}{360}$ of the whole circumference $2\pi r$:

$$s = \frac{40}{360} \cdot 2\pi(9) = \frac{1}{9} \cdot 18\pi = 2\pi = 6.2832\ldots$$

So $\boxed{s \approx 6.28 \text{ cm}}$

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**4.** Chords $\overline{AB}$ and $\overline{CD}$ meet at $E$ inside the circle; $AE = 6$, $EB = 10$, $CE = 4$. Find $ED$.
**Solution:** The intersecting-chords theorem says the two products of the pieces are equal: $AE \cdot EB = CE \cdot ED$.

$$\begin{aligned} 6 \cdot 10 &= 4 \cdot ED \\ 60 &= 4 \cdot ED \\ ED &= 15 \end{aligned}$$

So $\boxed{ED = 15}$

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**5.** Fountain $(x-5)^2 + (y+2)^2 = 49$ in metres; tangent at the top of the fountain. (a) Radius. (b) $y$ at the tangent point.
**Solution (a):** The right-hand side of centre-radius form is $r^2$, so $r = \sqrt{49}$. $\boxed{r = 7 \text{ m}}$
**Solution (b):** The centre is $(5, -2)$. The point directly above the centre sits one radius higher, so its $y$-coordinate is $-2 + 7$. $\boxed{5}$
(A tangent is perpendicular to the radius at the point of contact, which is why the tangent path here runs horizontally.)

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**6.** Cyclic quadrilateral $ABCD$: $m\angle A = (2x+15)°$, $m\angle C = (3x+40)°$. (a) Find $x$. (b) Find $m\angle A$.
**Solution (a):** $\angle A$ and $\angle C$ are opposite angles of a quadrilateral inscribed in a circle, so they are supplementary.

$$\begin{aligned} (2x+15) + (3x+40) &= 180 \\ 5x + 55 &= 180 \\ 5x &= 125 \\ x &= 25 \end{aligned}$$

So $\boxed{x = 25}$
**Solution (b):** $m\angle A = 2(25) + 15 = 50 + 15$. $\boxed{65°}$
Check: $m\angle C = 3(25) + 40 = 115°$, and $65 + 115 = 180$ ✓

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**7.** Radius 12 cm, central angle 120°. Find the sector area.
**Solution:** A sector is the fraction $\frac{120}{360}$ of the whole area $\pi r^2$:

$$A = \frac{120}{360} \cdot \pi(12)^2 = \frac{1}{3} \cdot 144\pi = 48\pi = 150.7964\ldots$$

So $\boxed{A \approx 150.80\text{ cm}^2}$

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**8.** Tangent $PT = 12$, secant from $P$ with external part $PA = 8$. Find $PB$.

**Solution:** The tangent-secant theorem: the tangent length *squared* equals the product of the whole secant and its external part, $PT^2 = PA \cdot PB$.

$$\begin{aligned} 12^2 &= 8 \cdot PB \\ 144 &= 8 \cdot PB \\ PB &= 18 \end{aligned}$$

So $\boxed{PB = 18}$

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**9.** Radius 10 cm, minor arc $AB = 72°$. (a) Arc length. (b) Sector area.

**Solution (a):** $s = \frac{72}{360} \cdot 2\pi(10) = \frac{1}{5} \cdot 20\pi = 4\pi = 12.5664\ldots$, so $\boxed{s \approx 12.57\text{ cm}}$

**Solution (b):** $A = \frac{72}{360} \cdot \pi(10)^2 = \frac{1}{5} \cdot 100\pi = 20\pi = 62.8319\ldots$, so $\boxed{A \approx 62.83\text{ cm}^2}$

Note the two formulas share the same fraction $\frac{72}{360}$; only the whole-circle quantity changes.

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**10.** Circle $x^2 + y^2 - 10x + 4y - 71 = 0$. (a) Centre-radius form and radius. (b) Distance to a 16-unit chord. (c) Why no 24-unit chord?

**Solution (a):** Complete both squares. Half of $-10$ is $-5$ and $(-5)^2 = 25$; half of 4 is 2 and $2^2 = 4$.

$$\begin{aligned} (x^2 - 10x + 25) + (y^2 + 4y + 4) &= 71 + 25 + 4 \\ (x-5)^2 + (y+2)^2 &= 100 \end{aligned}$$

So $\boxed{(x-5)^2 + (y+2)^2 = 100}$

The centre is $(5, -2)$ and $r = \sqrt{100}$, giving $\boxed{r = 10}$

**Solution (b):** A radius drawn to the midpoint of a chord is perpendicular to it, so the radius, half the chord, and the centre-to-chord distance form a right triangle with the radius as hypotenuse. Half the chord is 8.

$$d = \sqrt{10^2 - 8^2} = \sqrt{100 - 64} = \sqrt{36}$$

So $\boxed{6}$

**Solution (c) — instructor-judged.** A model response: the longest chord of any circle is a diameter, and this circle's diameter is $2 \times 10 = 20$. A chord of 24 would have to be longer than the diameter, which is impossible. (Algebraically, the right triangle in part (b) would need $d = \sqrt{100 - 144}$, and no real distance has a negative square.)

*Full credit:* computes the diameter as 20 and states that no chord can exceed the diameter. *Half credit:* gives the diameter but never names it as the maximum. *No credit:* only observes that 24 is bigger than 10.